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Predefined-time sliding mode control for position and attitude of quadrotor with variable exponent coefficients

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This paper presents a comprehensive study on predefined-time sliding mode control applied to quadrotor systems with variable exponent coefficients. The main goal is to achieve efficient control with predefined-time convergence to the desired target. Novel predefined-time sliding mode control and predefined-time neural networks sliding model control algorithms are developed to adapt to the dynamics of variable coefficients and ensure system stability. The stability analysis confirms their capability to achieve predefined-time convergence. Extensive numerical simulations on diverse nonlinear systems validate the superiority of the proposed approach over conventional sliding mode control and neural networks control.

Keywords Predefined-time stability, Sliding mode control, Neural networks, Trajectory tracking

Quadrotor aerial vehicles (UAVs) have gained significant attention in recent years due to their versatility in applications such as surveillance, search and rescue, payload delivery, and environmental monitoring. The effectiveness of UAV control strategies has a direct impact on overall system performance[1]. The control of nonlinear systems has garnered significant attention in various engineering and scientific fields due to their widespread applications in modern technology[2–4]. Nonlinear systems are known for their complex dynamic behavior and the presence of uncertainties, posing significant challenges in control design. To tackle these complexities and achieve robust and efficient control, researchers have explored various control methodologies[5,6]. Among these, sliding mode control has emerged as a powerful technique that provides excellent performance, especially for nonlinear systems[7–9].

Sliding mode control involves the introduction of a sliding surface in the state space to guide the system states towards a desired target rapidly[10,11]. Its robustness against parameter variations and external disturbances makes it suitable for stabilizing complex systems[12–14]. However, conventional sliding mode control suffers from one limitation: the convergence time is typically unbounded, which may not be acceptable in time-critical applications such as autonomous vehicles or robotic systems[15–17].

Predefined-time (PT) control[18,19] is a novel control strategy that ensures convergence time in a strictly predefined settling time, independent of initial conditions. Unlike traditional control methods that linear feedback control only ensures asymptotic stability with infinite time, finite/fixed-time[20] and predefined-time control guarantee that the system state will reach the target point exactly at or before a specified time. To address this limitation and enhance the applicability of sliding mode control, a relatively new control technique known as fixed-time sliding mode control (FT-SMC). There exist some predefined-time control method, such as arbitrary convergence[21] and variable exponent coefficient[22,23] have been developed. For arbitrary convergence control, there exist result[21], comments[24,25] and authors' reply in[26], proposed sufficient condition for Lyapunov stability given for arbitrary chosen time stable system. For variable exponent coefficient (VEC)[22,23] and comment[27], it is hard to get the bounded of the convergence time under different initial conditions. FT-SMC imposes a predefined convergence time constraint, ensuring that the system reaches the desired target within a specified finite time[28]. This unique feature makes FT-SMC an ideal choice for real-time control applications, where precise and timely control responses are essential. For instance, the work in[29] proposes finite-time sliding mode control with mismatched disturbances, but the convergence time bound depends on system parameters and initial conditions. Predefined-time[30], prescribed-time[31] and our VEC method allows the user to predefine the

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convergence time directly as a tunable parameter. Predefined-time VEC-based design provides a rigorous and straightforward Lyapunov-based stability proof with a clear and computable settling time upper bound.

Nonetheless, real-world systems often exhibit variable exponent coefficients, introducing additional complexities in the control process. Examples of such systems include variable coefficient circuits, adaptive control systems, and time-varying systems. These variable exponent coefficients challenge the application of traditional FT-SMC methods, necessitating innovative solutions to tackle this class of control problems. Therefore, the primary focus of this research is to explore and apply FT-SMC to nonlinear systems with variable exponent coefficients. The objective is to develop an effective control strategy that can handle the challenges posed by such systems and achieve robust and predefined-time convergence to the desired set-point.

This paper presents a comprehensive investigation into the theoretical foundation of PT-SMC, analyzing its advantages and applicability in addressing nonlinear control problems. Subsequently, a novel PT-SMC algorithm will be derived, tailored specifically for nonlinear systems with variable exponent coefficients. The design principles and control strategies of the proposed algorithm will be thoroughly examined. Furthermore, the stability of the derived control algorithm will be rigorously analyzed using mathematical tools to prove its capability to achieve predefined-time convergence, independent of system uncertainties and initial conditions. Numerical simulations will be conducted on various nonlinear systems with variable exponent coefficients to validate the performance and effectiveness of the proposed PT-SMC approach. Comparisons with traditional sliding mode control methods will be presented to highlight the advantages of the proposed technique.

The outcome of this research is expected to provide an efficient control solution for nonlinear systems with variable exponent coefficients, showcasing the potential of PT-SMC in real-time applications. The developed methodology may find application in diverse fields, including power electronics, robotics, and automation, where fast and accurate control responses are crucial for achieving optimal system performance. Ultimately, this study aims to contribute to the advancement of control theory and promote the adoption of predefined-time sliding mode control in a wide range of engineering and scientific domains.

In comparison to existing fixed-time and predefined-time control schemes[18–20], the main novelties of this work are threefold: Firstly, the proposed Variable Exponent Coefficient (VEC) function provides a more flexible framework for tuning the predefined convergence time, overcoming the conservatism of fixed-exponent methods. Secondly, the integration of VEC with both standard sliding mode control and neural network-based adaptive control offers a unified approach to handle system uncertainties and nonlinearities with a guaranteed settling time. Thirdly, the application to the full six-degree-of-freedom quadrotor model, including both position and attitude tracking, demonstrates the practical viability of the method in complex, underactuated systems, which is not fully explored in the aforementioned works.

The main contributions are summarized as following.

(1) The predefined-time sliding mode control method is proposed by variable exponent coefficients. Neural networks sliding mode control is designed for quadrotor dynamic system, and the proposed control strategy guarantees that the PT stable.
(2) Based on neural networks sliding mode control and Lyapunov theory, the close-loop system can be proved PT stable. The variable exponent coefficients adaptive law is proposed for neural networks.
(3) The sliding mode PT control scheme is proposed for quadrotor, which refers to a control strategy where the system reaches the desired state within a predefined and finite time, regardless of initial conditions. This approach ensures convergence in a fixed time frame, enhancing predictability and robustness in dynamic systems.

The rest of the paper consists of the following parts. In "System description and problem formulation" section, mathematical description of quadrotor dynamic system and some basic definitions are presented. In "Sliding mode control with VEC" section, the sliding mode PT control with VEC is presented. In "Neural Networks sliding mode control with VEC" section, the neural networks sliding mode control is proposed for quadrotor dynamic system is proposed. In "Simulation example" section, sliding mode control and quadrotor dynamic system neural networks control results are illustrated by simulation examples. In "Conclusion" section, a conclusion is summarized.

System description and problem formulation

In this section, the system description, predefined-time control and preliminary are present. The definition of key symbols and their units is in Table 1.

Symbol	Description	Unit
(x, y, z)	Position in inertial frame	m
(ϕ, θ, ψ)	Roll, pitch, yaw angles	rad
m	Mass of the quadrotor	kg
I_{xx}, I_{yy}, I_{zz}	Moments of inertia	kg·m^2
U_1, U_2, U_3, U_4	Control inputs	N, N m

Table 1. Definition of key symbols and their units.

System description

The quadrotor dynamic model is described via Newton–Euler equations as

$$\begin{cases} \ddot{x} = \frac{u_1}{m}(\sin\psi\sin\phi + \cos\psi\sin\theta\cos\phi) - \frac{p_x}{m}\dot{x} \\ \ddot{y} = \frac{u_1}{m}(-\cos\psi\sin\phi + \sin\psi\sin\theta\cos\phi) - \frac{p_y}{m}\dot{y} \\ \ddot{z} = \frac{u_1}{m}(\cos\theta\cos\phi) - g - \frac{p_z}{m}\dot{z} \\ \ddot{\phi} = \dot{\theta}\dot{\psi}\frac{J_y - J_z}{J_x} - \frac{I_r}{J_x}\dot{\theta}\varpi - \frac{p_\phi}{J_x}\dot{\phi} + \frac{u_2}{J_x} \\ \ddot{\theta} = \dot{\phi}\dot{\psi}\frac{J_z - J_x}{J_y} - \frac{I_r}{J_y}\dot{\phi}\varpi - \frac{p_\theta}{J_y}\dot{\theta} + \frac{u_3}{J_y} \\ \ddot{\psi} = \dot{\theta}\dot{\phi}\frac{J_x - J_y}{J_z} - \frac{p_\psi}{J_z}\dot{\psi} + \frac{u_4}{J_z} \end{cases} \tag{1}$$

where $[\ x \quad y \quad z\ ]^T$ indicate Euclidean position and $[\ \phi \quad \theta \quad \psi\ ]^T$ indicate Euler angles roll, pitch, yaw of quadrotor. m, J_x, J_y, J_z indicate the mass and the moment of inertia in the three directions, respectively. $p_x, p_y, p_z, p_\phi, p_\theta, p_\psi$ indicate the aerodynamic damping coefficients. I_r indicate the rotor inertia, ϖ is the overall remaining rotor angle as $\varpi = \omega_4 + \omega_3 - \omega_1 - \omega_2$, where ω_i is given in terms of angular speed of the i th rotor. u_i indicate the control input and g is gravity acceleration.

The virtual control input is designed as

$$\begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} = \begin{bmatrix} (\sin\psi\sin\phi + \cos\psi\sin\theta\cos\phi)u_1 \\ (-\cos\psi\sin\phi + \sin\psi\sin\theta\cos\phi)\,u_1 \\ (\cos\theta\cos\phi)u_1 \end{bmatrix} \tag{2}$$

then, design the control laws u_x, u_y, u_z, u_4 for translational system and $u_1 = \sqrt{u_x^2 + u_y^2 + u_z^2}$. With variable exponent coefficients, the predefined-time sliding mode control is designed to for the quadrotor to track the reference trajectory such as:

$$\lim_{t\to T_c} x \to x_d,\ \lim_{t\to T_c} y \to y_d,\ \lim_{t\to T_c} z \to z_d,\ \lim_{t\to T_c} \psi \to \psi_d \tag{3}$$

therefore

$$\begin{cases} \ddot{\chi}_x = -\frac{p_x}{m}\dot{\chi}_x - \frac{p_x}{m}\dot{x}_d - \ddot{x}_d + \frac{u_x}{m} \\ \ddot{\chi}_y = -\frac{p_y}{m}\dot{\chi}_y - \frac{p_y}{m}\dot{y}_d - \ddot{y}_d + \frac{u_y}{m} \\ \ddot{\chi}_z = -\frac{p_z}{m}\dot{\chi}_z - \frac{p_z}{m}\dot{z}_d - g - \ddot{z}_d + \frac{u_z}{m} \end{cases} \tag{4}$$

where $\chi_x = x - x_d$, $\chi_y = y - y_d$, $\chi_z = z - z_d$, then get desired roll and pitch references from

$$\phi_d = \arcsin\left(\frac{u_x\sin(\psi_d) - u_y\cos(\psi_d)}{u_1}\right) \tag{5}$$

$$\theta_d = \arctan\left(\frac{u_x\cos(\psi_d) + u_y\sin(\psi_d)}{u_z}\right) \tag{6}$$

then, $\chi_\phi = \phi - \phi_d,\ \chi_\theta = \theta - \theta_d, \chi_\psi = \psi - \psi_d$

, therefore,

$$\begin{cases} \ddot{\chi}_\phi = -\frac{p_\phi}{J_x}\dot{\chi}_\phi - \frac{p_\phi}{J_x}\dot{\phi}_d + \left(\dot{\chi}_\theta + \dot{\theta}_d\right)\dot{\psi}\frac{J_y - J_z}{J_x} - \frac{I_r}{J_x}\left(\dot{\chi}_\theta + \dot{\theta}_d\right)\varpi - \ddot{\phi}_d + \frac{u_2}{J_x} \\ \ddot{\chi}_\theta = -\frac{p_\theta}{J_y}\dot{\chi}_\theta - \frac{p_\theta}{J_y}\dot{\theta}_d + \left(\dot{\chi}_\phi + \dot{\phi}_d\right)\dot{\psi}\frac{J_z - J_x}{J_y} - \frac{I_r}{J_y}\left(\dot{\chi}_\phi + \dot{\phi}_d\right)\varpi - \ddot{\theta}_d + \frac{u_3}{J_y} \\ \ddot{\chi}_\psi = -\frac{p_\psi}{J_z}\dot{\chi}_\psi - \frac{p_\psi}{J_z}\dot{\psi}_d + \left(\dot{\chi}_\theta + \dot{\theta}_d\right)\left(\dot{\chi}_\phi + \dot{\phi}_d\right)\frac{J_x - J_y}{J_z} - \ddot{\psi}_d + \frac{u_4}{J_z} \end{cases} \tag{7}$$

therefore, the accurate trajectory tracking problem is changed as

$$\lim_{t\to T_c} \chi_x \to 0,\ \lim_{t\to T_c} \chi_y \to 0,\ \lim_{t\to T_c} \chi_z \to 0,\ \lim_{t\to T_c} \chi_\psi \to 0 \tag{8}$$

Then, the quadrotor dynamic systems trajectory tracking control is 2-order system.

Predefined time control

Consider the following nonlinear system:

$$\dot{x} = f(t, x) \tag{9}$$

where $f : [0, \infty) \times \mathbb{R}^n \to \mathbb{R}$ represents a nonlinear continuous function, and initial condition $x_0 = x(0)$.

Definition 1[28] System (9) is predefined time stable if there exist a positive instant T_c, then the solution $x(t, x_0)$ is stable at T_c, where $T_c = \inf\{t_c(x_0)\}$ and

$$t_c(x_0) = \inf\{t | x(t, x_0) \equiv 0, t \geq t_c\} \tag{10}$$

where T_c is set a priori by the user as parameter.

Lemma 1 (Variable Exponent Coefficient) Suppose the continuous radically unbounded function $V(\cdot)\ R^n \to R_+ \cup \{0\}$ such that.

$$\dot{V} \leq -\frac{\Pi}{T_c} V^{\varphi(V)} \tag{11}$$

where $\varphi(V) = \frac{bV+a}{aV+b}$, $a > b > 0$, $0 < \alpha < 1$, $\beta = \frac{b(1+\alpha)+a}{a(1+\alpha)+b}$, $\gamma = \frac{b(1-\alpha)+a}{a(1-\alpha)+b}$ and

$$\Pi = \max_{0<\alpha<1} \left(\frac{(1+\alpha)^{1-\beta}}{\beta - 1} + \frac{2\alpha}{(1-\alpha)^{\varphi(1-\alpha)}} + \frac{(1-\alpha)^{1-\gamma}}{1-\gamma} \right) \tag{12}$$

then the system is predefined-time stable, and settling time is $T = \sup_{x_0 \in \Re^n} T(x_0) \leq T_c$.

Proof Without loss of generality, assume that $V(0) = V_0 > 1$ and positive instant $V_0 > 1 + \alpha > 1$. Because $\varphi(V) = \frac{bV+a}{aV+b} > \frac{b}{a}$ and Eq. (11), the solution $V(t)$ is stable.

First step, V from V_0 to $1 + \alpha$ in finite time as T_1. Assume the $\beta = \frac{b(1+\alpha)+a}{a(1+\alpha)+b}$, then for dynamic system (11), then $\dot{V} \leq -kV^{\varphi(V)} \leq -kV^{\beta}$, therefore

$$\frac{dV}{V^{\beta}} = -kdt \tag{13}$$

integrate Eq. (13)

$$\frac{1}{1-\beta} V^{1-\beta} = -kt + C \tag{14}$$

where parameter C can be get through initial conditions $t = 0$ and V_0, then

$$C = \frac{1}{1-\beta} V_0^{1-\beta} \tag{15}$$

therefore

$$\frac{1}{1-\beta} V^{1-\beta} = -kt + \frac{1}{1-\beta} V_0^{1-\beta} \tag{16}$$

then

$$kt = \frac{1}{\beta - 1} V^{1-\beta} - \frac{1}{\beta - 1} V_0^{1-\beta} \tag{17}$$

and

$$\begin{aligned} T_1 &= \frac{1}{k(\beta-1)} (1+\alpha)^{1-\beta} - \frac{1}{k(\beta-1)} V_0^{1-\beta} \\ &< \frac{1}{k(\beta-1)} (1+\alpha)^{1-\beta} \end{aligned} \tag{18}$$

Therefore, the V from V_0 to $1 + \alpha$ in finite time as T_1.

Second step, in $V \in (1-\alpha, 1+\alpha)$, $V^{\varphi(V)}$ is a monotonically increasing function, therefore $V^{\varphi(V)} \geq (1-\alpha)^{\varphi(1-\alpha)}$, and

$$\frac{dV}{dt} \leq -k(1-\alpha)^{\varphi(1-\alpha)} \tag{19}$$

then

$$V(t) = -k(1-\alpha)^{\varphi(1-\alpha)} t + 1 + \alpha \tag{20}$$

then

$$T_2 = \frac{2\alpha}{k(1-\alpha)^{\varphi(1-\alpha)}} \tag{21}$$

Therefore, the V from $1+\alpha$ to $1-\alpha$ in finite time as T_2.

Third step $\varphi(V) = \frac{bV+a}{aV+b}$, $\gamma = \frac{b(1-\alpha)+a}{a(1-\alpha)+b}$, then

$$\frac{dV}{dt} = -kV^{\gamma} \tag{22}$$

Integrate the above equation to obtain

$$V(t) = \left(-kt(1-\gamma) + (1-\alpha)^{1-\gamma}\right)^{\frac{1}{1-\gamma}} \tag{23}$$

$$T_3 = \frac{1}{k(1-\gamma)}(1-\alpha)^{1-\gamma} \tag{24}$$

Remark 1 For Π in (12) is minimum when $0 < \alpha < 1$, the minimum can be calculated easily after choose parameters a, b, k, and it is independent of the initial conditions.

Preliminaries

Lemma 2 The two vectors $X = [x_1, \ldots, x_n]^T$, $Y = [y_1, \ldots, y_n]^T$, then the following inequation holds:

$$\begin{aligned} |x_1y_1 + x_2y_2 + \cdots + x_ny_n| &\le \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}\sqrt{y_1^2 + y_2^2 + \cdots + y_n^2} \\ &\le \|X\| \, \|Y\| \end{aligned} \tag{25}$$

Lemma 3 The $\hat{\theta} > 0, \theta > 0$ and $a > b > 0$, then the following inequation holds:

$$\tilde{\theta}\hat{\theta}^{\frac{b+a\hat{\theta}^2}{a+b\hat{\theta}^2}} \le -\lambda\left|\tilde{\theta}\right|^{\frac{2b+2a\tilde{\theta}^2}{a+b\hat{\theta}^2}} + \mu\theta^{\frac{2b+2a\theta^2}{a+b\theta^2}} \tag{26}$$

where $\lambda > 0$,$\mu > 0$, $\tilde{\theta} = \hat{\theta} - \theta$.

Lemma 4 The $a > b > 0$,$V_i > 0, i = 1, 2, \ldots, n$, the following inequation hold:

$$V_1^{\frac{aV_1+b}{a+bV_1}} + V_2^{\frac{aV_2+b}{a+bV_2}} + \cdots + V_n^{\frac{aV_n+b}{a+bV_n}} \ge \vartheta(V_1 + V_2 + \cdots + V_n)^{\frac{a(V_1+V_2+\cdots+V_n)+b}{a+b(V_1+V_2+\cdots+V_n)}} \tag{27}$$

where $\vartheta > 0$.

Finally, the symbol $\lceil x \rfloor^{\alpha} = |x|^{\alpha} \, sign(x)$ is determined for $\alpha \in R$, and $\lceil x \rfloor^0 = sign(x)$.

Sliding mode control with VEC

For nonlinear system

$$\ddot{x} = f(x, \dot{x}) + gu \tag{28}$$

definite the $x_1 = x, x_2 = \dot{x}$, then

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x_1, x_2) + gu \end{cases} \tag{29}$$

design the sliding mode as

$$s = x_2 + \frac{k_1}{2}\lceil x_1 \rfloor^{\frac{2bx_1^2+2a}{ax_1^2+b}-1} \tag{30}$$

If $s = 0$, $x_2 = -\frac{k_1}{2}\lceil x_1 \rfloor^{\frac{2bx_1^2+2a}{ax_1^2+b}-1}$, therefore

$$\dot{x}_1 = -\frac{k_1}{2}\lceil x_1 \rfloor^{\frac{2bx_1^2+2a}{ax_1^2+b}-1} \tag{31}$$

then

$$\dot{s} = \dot{x}_2 + \frac{k_1}{2} d \tag{32}$$

where

$$\begin{aligned} d = &- \frac{2\left|x_1\right| \ln\left(\left|x_1\right|\right) \left(\frac{a-2b}{a x_1^2+b} - \frac{a\left((a-2b) x_1^2-2a+b\right)}{\left(a x_1^2+b\right)^2}\right)}{\left|x_1\right|^{\frac{(a-2b) x_1^2-2a+b}{a x_1^2+b}}} \\ &- \frac{(a-2b) x_1^2 - 2a + b}{\left|x_1\right|^{\frac{(a-2b) x_1^2-2a+b}{a x_1^2+b}+1} \left(a x_1^2 + b\right)} \end{aligned} \tag{33}$$

then

$$\dot{s} = f(x) + gu + \frac{k_1}{2} d \tag{34}$$

Design the control as

$$u = -\frac{1}{g}\left(\frac{k_2}{2}\lceil s\rfloor^{\frac{2bs^2+2a}{as^2+b}-1} + f(x) + \frac{k_1}{2} d\right) \tag{35}$$

Theorem 1 Given the nonlinear system (28) with sliding mode (30) and application of controller (35), the error signal x converges to zero with predefined time.

Proof Choose the Lyapunov candidate functional as.

$$V_1 = x_1^2 \tag{36}$$

Differentiating (36) along the system trajectories yields

$$\begin{aligned} \dot{V}_1 =& - k_1 x_1 \lceil x_1 \rfloor^{\frac{2bx_1^2+2a}{ax_1^2+b}-1} \\ =& - k_1 V_1^{\frac{bV_1+a}{aV_1+b}} \end{aligned} \tag{37}$$

Then

$$\dot{s} = -\frac{k_2}{2}\lceil s\rfloor^{\frac{2bs^2+2a}{as^2+b}-1} \tag{38}$$

Choose the Lyapunov candidate functional as

$$V_2 = s^2 \tag{39}$$

Differentiating (39) along the system trajectories yields

$$\begin{aligned} \dot{V}_2 =& 2s\dot{s} \\ =& - k_2 V_2^{\frac{bV_2+a}{aV_2+b}} \end{aligned} \tag{40}$$

Therefore, based on the Lemma 1 the close-loop system is predefined-time stability. □

Neural networks sliding mode control with VEC

In this section, the main result of Theorem 1 is extended to the neural networks control, then

$$\dot{s} = f(x) + gu + \frac{k_1}{2} d \tag{41}$$

Choose the neural networks

$$\begin{aligned} f(x) + \frac{k_1}{2} d =& W^T \Psi(\overline{x}_2) + \varepsilon \\ \leq& \theta \left\|\Psi(\overline{x}_2)\right\| + \varepsilon \end{aligned} \tag{42}$$

where $\theta = \|W\|$, then the controller can be designed as

$$u = -\frac{1}{g}\left(\frac{k_2}{2}\lceil s\rfloor^{\frac{2bs^2+2a}{as^2+b}-1} + \frac{k_3}{2}s + \lceil s\rfloor^0\hat{\theta}\left\|\Psi\left(\overline{x}_2\right)\right\| + \frac{\overline{\delta}}{2\lceil s\rfloor}\right) \tag{43}$$

where $k_2 > 0, k_3 > 0$, and the adaptive law is choose as

$$\dot{\hat{\theta}} = |s|\left\|\Psi\left(\overline{x}_2\right)\right\| - \frac{\varsigma}{2}\hat{\theta}^{\frac{b+a\hat{\theta}^2}{a+b\hat{\theta}^2}}, \hat{\theta}\left(0\right) = 0 \tag{44}$$

Theorem 2 Given the nonlinear system (28) application of neural networks controller (43) and adaptive law as (44), the error signal x converges to zero with predefined time.

Proof Select the Lyapunov functional as.

$$V_2 = s^2 + \tilde{\theta}^2 \tag{45}$$

Differentiating (45) along the system trajectories yields

$$\begin{aligned}\dot{V}_2 =& 2s\dot{s} + 2\tilde{\theta}\dot{\hat{\theta}} \\ \leq& 2s\left(\theta\left\|\Psi\left(\overline{x}_2\right)\right\| + \varepsilon + gu\right) + 2\tilde{\theta}\left|s\right|\left\|\Psi\left(\overline{x}_2\right)\right\| - 2\varsigma\tilde{\theta}\hat{\theta}^{\frac{b+a\hat{\theta}^2}{a+b\hat{\theta}^2}}\end{aligned} \tag{46}$$

Designed the controller and neural networks adaptive law

$$\begin{aligned}\dot{V}_2 \leq& 2s\left(\theta\left\|\Psi\left(\overline{x}_2\right)\right\| + \varepsilon\right) - 2s\left(\frac{k_2}{2}\lceil s\rfloor^{\frac{2bs^2+2a}{as^2+b}-1} + \frac{k_3}{2}s + \lceil s\rfloor^0\hat{\theta}\left\|\Psi\left(\overline{x}_2\right)\right\| + s^{-1}\frac{\overline{\delta}}{2}\right) \\ &+ 2\tilde{\theta}\left|s\right|\left\|\Psi\left(\overline{x}_2\right)\right\| - 2\varsigma\tilde{\theta}\hat{\theta}^{\frac{b+a\hat{\theta}^2}{a+b\hat{\theta}^2}} \\ \leq& 2s\varepsilon - k_2\left|s\right|^{\frac{2bs^2+2a}{as^2+b}} - k_3s^2 - 2\varsigma\tilde{\theta}\hat{\theta}^{\frac{b+a\hat{\theta}^2}{a+b\hat{\theta}^2}} - \overline{\delta}\end{aligned} \tag{47}$$

Based on inequation

$$-k_3s^2 + 2s\varepsilon \leq \frac{\varepsilon^2}{k_3} \tag{48}$$

and

$$-\varsigma\tilde{\theta}\hat{\theta}^{\frac{b+a\hat{\theta}^2}{a+b\hat{\theta}^2}} \leq -\lambda\left|\tilde{\theta}\right|^{\frac{2b+2a\tilde{\theta}^2}{a+b\hat{\theta}^2}} + \mu\theta^{\frac{2b+2a\theta^2}{a+b\theta^2}} \tag{49}$$

Choose $\overline{\delta} > \frac{\varepsilon^2}{k_3} + 2\mu\theta^{\frac{2b+2a\theta^2}{a+b\theta^2}}$, then based on Lemma 4

$$\begin{aligned}\dot{V}_2 \leq& -k_2\left(s^2\right)^{\frac{as^2+b}{a+bs^2}} - 2\lambda\left(\tilde{\theta}^2\right)^{\frac{a\hat{\theta}^2+b}{a+b\hat{\theta}^2}} \\ \leq& -k_4V_2^{\frac{aV_2+b}{a+bV_2}}\end{aligned} \tag{50}$$

Therefore, based on the Lemma 1 the close-loop system is predefined-time stability. □

Figure 1 is a diagram of the correlation between core terms of sliding mode control and neural network control, and Fig. 2 is the core formulas of sliding mode control and neural network control with VEC. These two figures list the key terms in the predefined-time sliding mode control (including variable exponent coefficients VEC) and neural network control studied in this paper, including performance indicators and their derivatives, transformation errors, virtual control inputs, adaptive laws, actual control inputs, neural network approximation, uncertain terms, sliding mode surfaces, etc. They intuitively present the correlation between various core elements in the design of control strategies.

Simulation example

To verify the effectiveness of the sliding mode control for position and attitude of quadrotor based on VEC, mathematical model and quadrotor model simulations are taken for performance analysis.

Simulation experiment of mathematical model

Consider the nonlinear system as:

$$\ddot{x} = f\left(x, \dot{x}\right) + u \tag{51}$$

define the $x_1 = x, x_2 = \dot{x}$, then

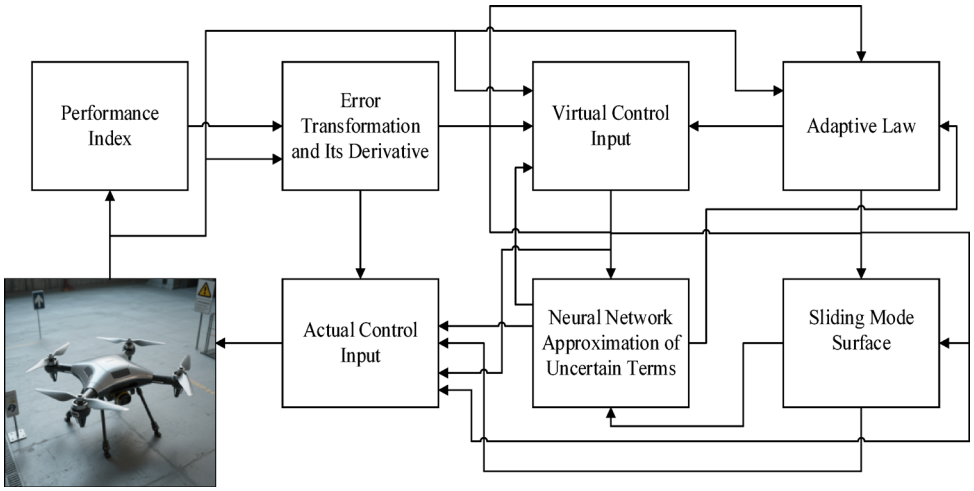

Fig. 1. Schematic diagram of adaptive sliding mode control architecture.

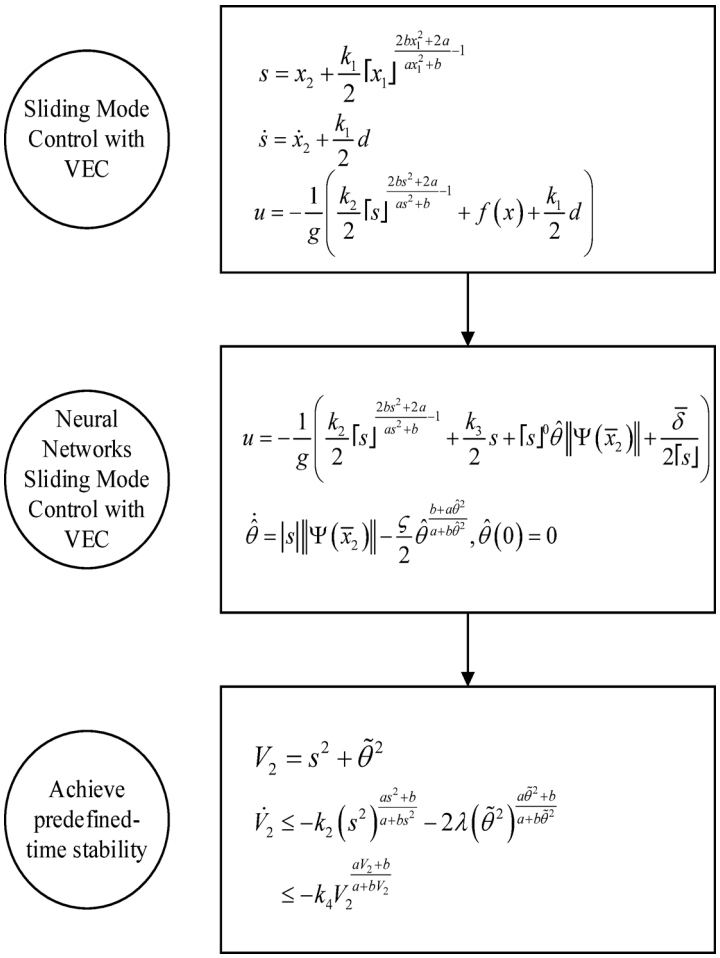

Fig. 2. VEC sliding mode control and neural network control algorithms.

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_1 \sin(x_2) + u \end{cases} \tag{52}$$

Based on sliding mode control in theorem 1, the parameters are set as $k_1 = 4, k_2 = 4, a = 7, b = 9$. The initial values of the system are set as $[x_1(0), x_2(0)]^T = [10, 0]^T$.

Design the sliding surface as

$$s = x_2 + 2\lceil x_1 \rfloor^{\frac{11x_1^2+5}{7x_1^2+9}} \tag{53}$$

and controller as

$$u = -2\lceil s \rfloor^{\frac{11s^2+5}{7s^2+9}} - x_1 \sin(x_2) - 2d \tag{54}$$

where $d = -\frac{2|x_1|\ln(|x_1|)\left(\frac{-11}{7x_1^2+9} - \frac{7(-11x_1^2-5)}{(7x_1^2+9)^2}\right)}{\frac{-11x_1^2-5}{7x_1^2+9}} - \frac{-11x_1^2-5}{\frac{-11x_1^2-5}{7x_1^2+9}+1}$.

Based on parameters $k_1 = 4, k_2 = 4, a = 7, b = 9$ and settling time for $0 < \alpha < 1$, the minimum settling time can be calculated as $T_{\max} = 4.3854$. The simulation results are shown in Figs. 3, 4, 5.

Note that the theoretical upper bound for the settling time is $T_{\max} = 4.3854$ s, as guaranteed by our stability analysis under worst-case conditions. However, it is shows that the states converge visually within approximately 1.5 s. This divergence is not only anticipated but also underscores a principal advantage—Lyapunov-based theoretical bounds are often conservative by nature. In practice, convergence occurs more rapidly and is contingent upon the initial conditions and the selected parameters of the VEC function. The guarantee is that convergence will occur no later than $T_{\max}$, which is user-defined.

From the above simulation results, the states x_1, x_2 are clearly stability with predefined time $T_{\max}$ in Fig. 3, sliding manifold is shown in Fig. 4 and sliding mode controller is shown in Fig. 5.

Remark 2 It can be observed from Fig. 5 that the control input values are large. This is a common characteristic of predefined-time and fixed-time controllers, which utilize high-gain feedback to achieve rapid convergence within a strict time frame. While effective in simulation, practical implementation must consider actuator saturation. This issue can be effectively addressed by employing saturation limits or low-pass filters within the control algorithm.

Simulation experiment of quadrotor model

The quadrotor dynamic model is described via Newton–Euler equations as

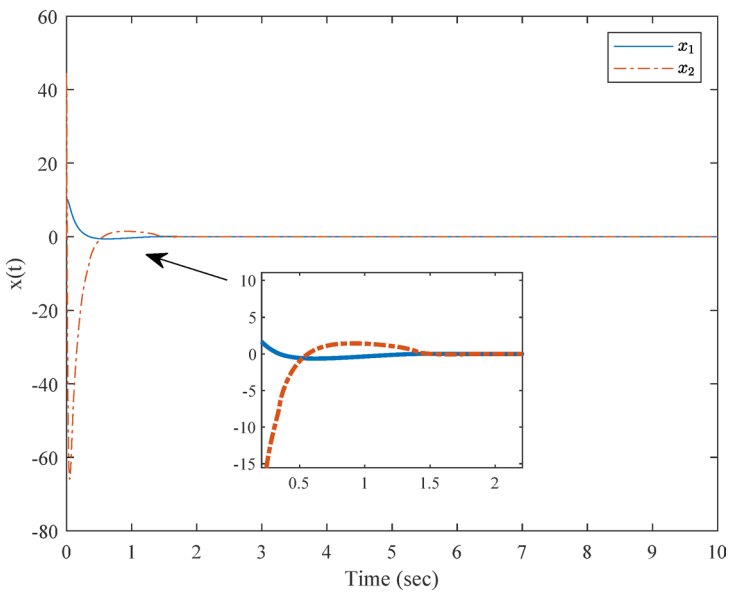

Fig. 3. The trajectory of x_1 and x_2.

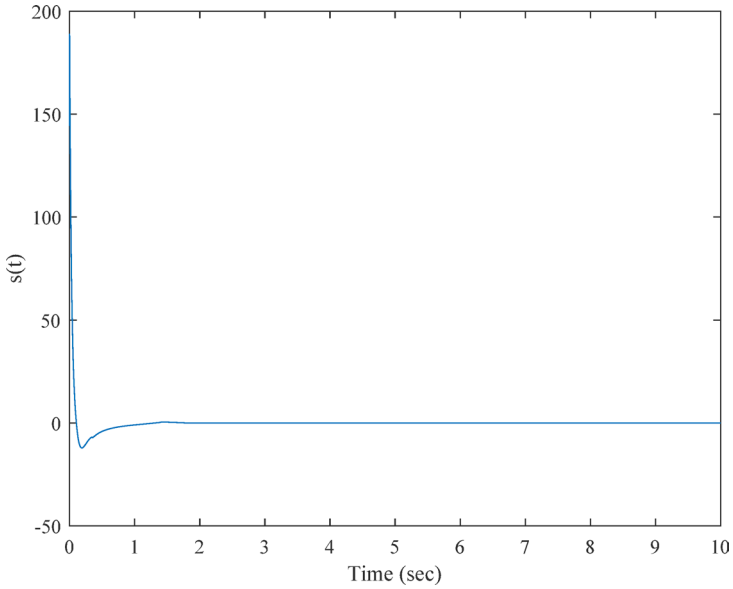

Fig. 4. Sliding manifold.

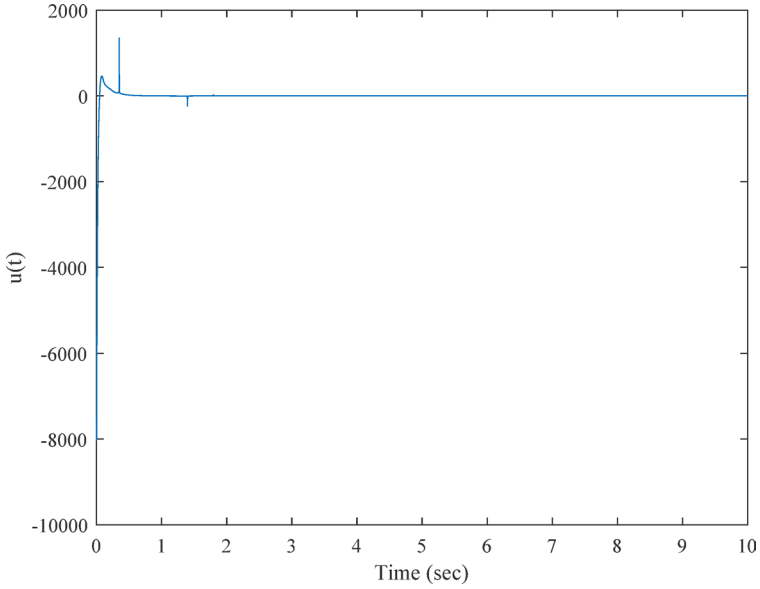

Fig. 5. System control of the nonlinear system.

$$\begin{cases} \ddot{x} = \frac{u_1}{m}(\sin\psi\sin\phi + \cos\psi\sin\theta\cos\phi) - \frac{p_x}{m}\dot{x} \\ \ddot{y} = \frac{u_1}{m}(-\cos\psi\sin\phi + \sin\psi\sin\theta\cos\phi) - \frac{p_y}{m}\dot{y} \\ \ddot{z} = \frac{u_1}{m}(\cos\theta\cos\phi) - g - \frac{p_z}{m}\dot{z} \\ \ddot{\phi} = \dot{\theta}\dot{\psi}\frac{J_y - J_z}{J_x} - \frac{I_r}{J_x}\dot{\theta}\varpi - \frac{p_\phi}{J_x}\dot{\phi} + \frac{u_2}{J_x} \\ \ddot{\theta} = \dot{\phi}\dot{\psi}\frac{J_z - J_x}{J_y} - \frac{I_r}{J_y}\dot{\phi}\varpi - \frac{p_\theta}{J_y}\dot{\theta} + \frac{u_3}{J_y} \\ \ddot{\psi} = \dot{\theta}\dot{\phi}\frac{J_x - J_y}{J_z} - \frac{p_\psi}{J_z}\dot{\psi} + \frac{u_4}{J_z} \end{cases} \tag{55}$$

where $[\ x \quad y \quad z\ ]^T$ indicate Euclidean position and $[\ \phi \quad \theta \quad \psi\ ]^T$ indicate Euler angles roll, pitch, yaw of quadrotor. m, J_x, J_y, J_z indicate the mass and the moment of inertia in the three directions, respectively. $p_x, p_y, p_z, p_\phi, p_\theta, p_\psi$ indicate the aerodynamic damping coefficients. I_r indicate the rotor inertia, ϖ is the overall remaining rotor angle as $\varpi = \omega_4 + \omega_3 - \omega_1 - \omega_2$, where ω_i is given in terms of angular speed of the i th rotor. u_i indicate the control input and g is gravity acceleration. Based on parameters $k_1 = 4, k_2 = 4, a = 7, b = 9$ and settling time for $0 < \alpha < 1$, the minimum settling time can be calculated as $T_{\max} = 4.3854$. The simulation results are shown in Figs. 4, 5, 6, 7, 8, 9 and 10.

From the above simulation results, the position states x, y, z and angles ϕ, θ, ψ of quadrotor are converge to x_d, y_d, z_d and ϕ_d, θ_d, ψ_d, respectively. The error states $\chi_x, \chi_y, \chi_z, \chi_\phi, \chi_\theta, \chi_\psi$ are converge to origin in Figs. 6, 7, 8, 9, 10, 11. Figure 12 show the 3-D spiral trajectory of the quadrotor. It can be seen that from the simulation results that the trajectory of the quadrotor follows the predetermined path within the predefined-time.

Comparative analysis

To further validate the superiority of the proposed Predefined-Time SMC (PT-SMC) with VEC, comparative simulations with Model-free Control[32] and Fixed-Time Control[33] are conducted on the quadrotor model in Table 2. The performance is evaluated using Integral of Time multiplied by Absolute Error (ITAE), Integral of Squared Error (ISE), and Total Control Effort.

Conclusion

This paper has proposed a novel predefined-time sliding mode control strategy utilizing Variable Exponent Coefficients (VEC) for the trajectory tracking of quadrotor UAVs. The primary contributions of this work are: (1) The development of a VEC-based sliding manifold that ensures predefined-time convergence regardless of initial conditions; (2) The design of an adaptive neural network controller integrated with the VEC framework to compensate for complex uncertainties without compromising the predefined-time stability property; (3) The successful application and validation of the proposed control scheme on a high-fidelity, nonlinear quadrotor model, demonstrating superior performance in tracking complex 3D trajectories.

The application of FT-SMC to nonlinear systems with VEC proves to be an effective and robust control strategy. The developed algorithm with VEC demonstrates remarkable adaptability in handling predefined-

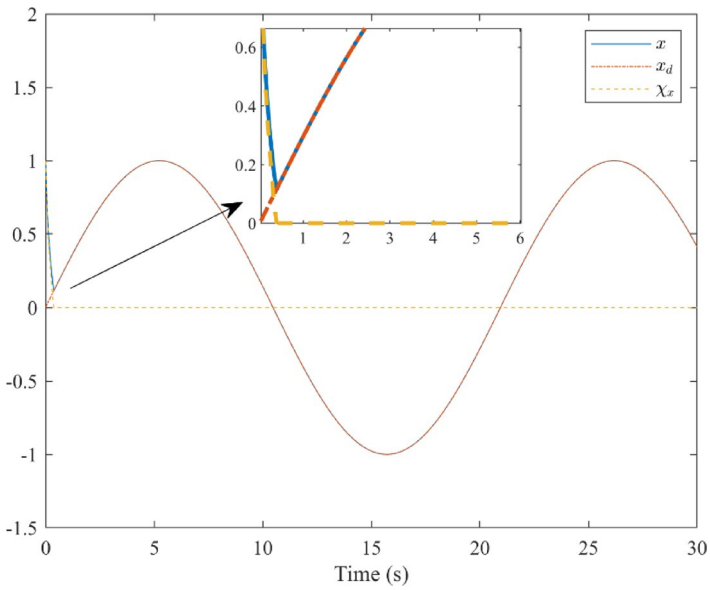

Fig. 6. The trajectory of x, x_d, χ_x.

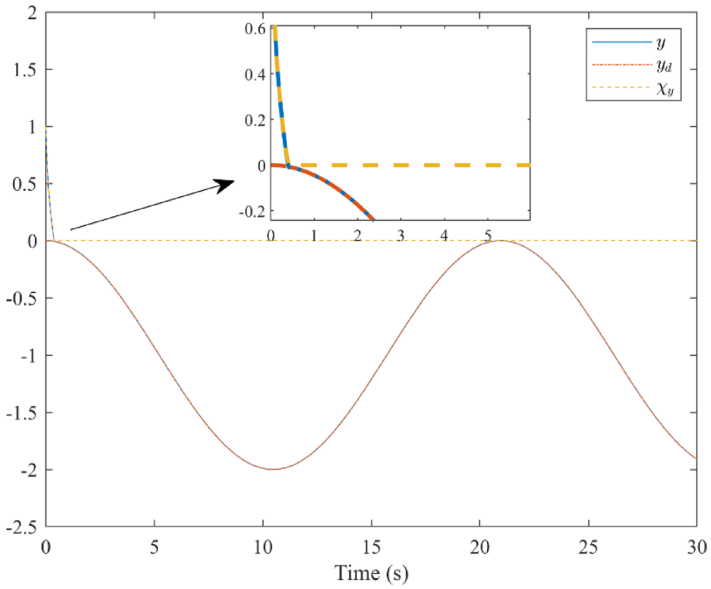

Fig. 7. The trajectory of y, y_d, χ_y.

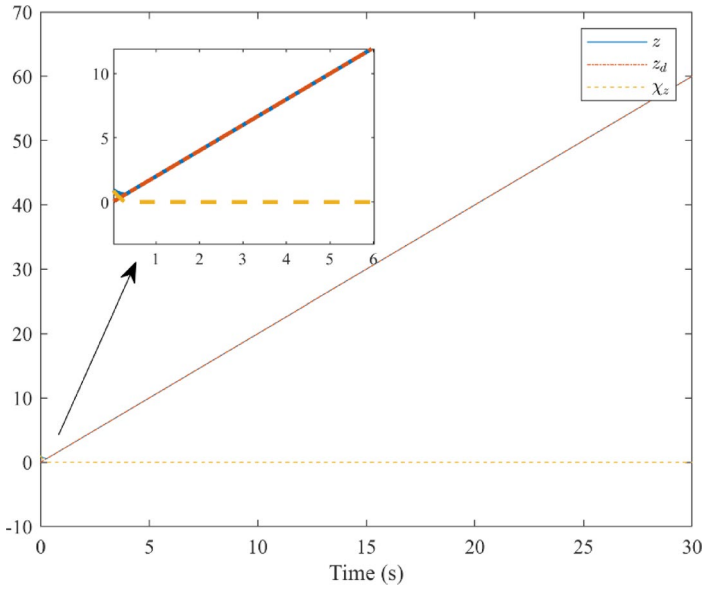

Fig. 8. The trajectory of z, z_d, χ_z.

time convergence, ensuring system stability, and providing reliable control performance. The stability analysis confirms its capability to achieve predefined-time convergence, independent of initial condition. Numerical simulations validate the superiority of FT-SMC over conventional sliding mode control, emphasizing its potential in time-critical applications. The fixed-time convergence property makes these control strategies suitable for time-critical applications, such as robotics, autonomous systems, and power electronics. This research advances control theory, offering a promising solution for real-world scenarios where precise and rapid control responses are essential for optimal performance.

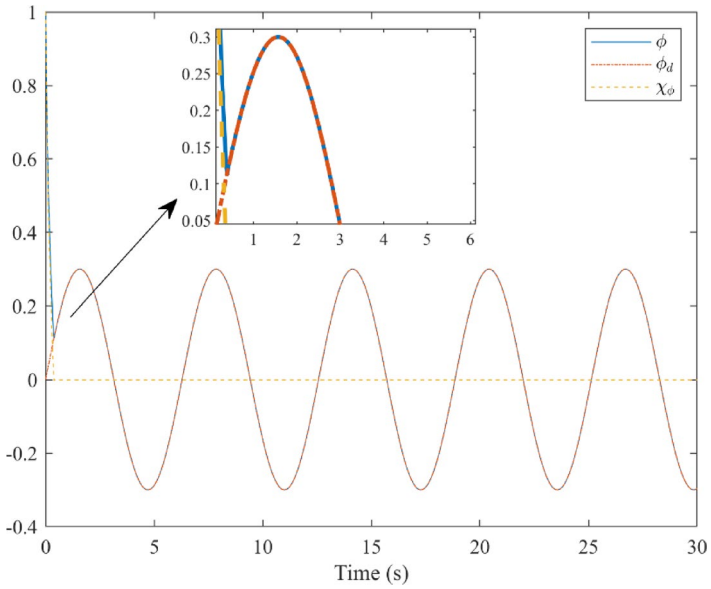

Fig. 9. The trajectory of ϕ, ϕ_d, χ_ϕ.

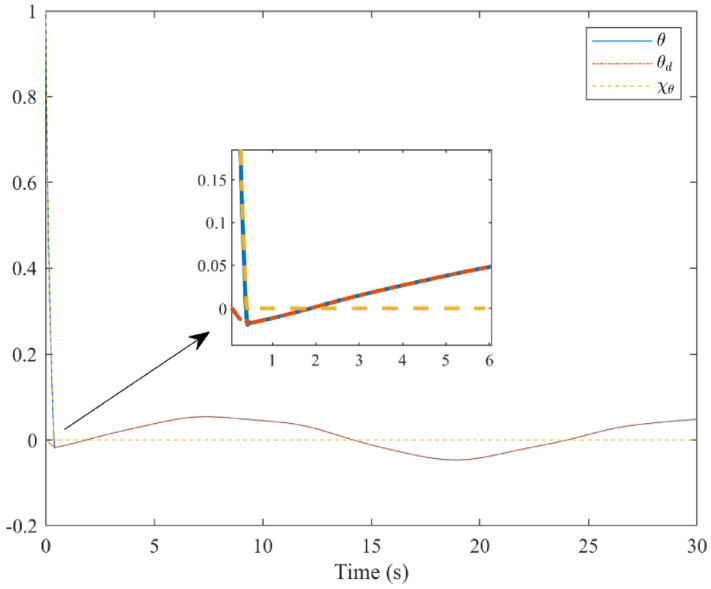

Fig. 10. The trajectory of $\theta, \theta_d, \chi_\theta$.

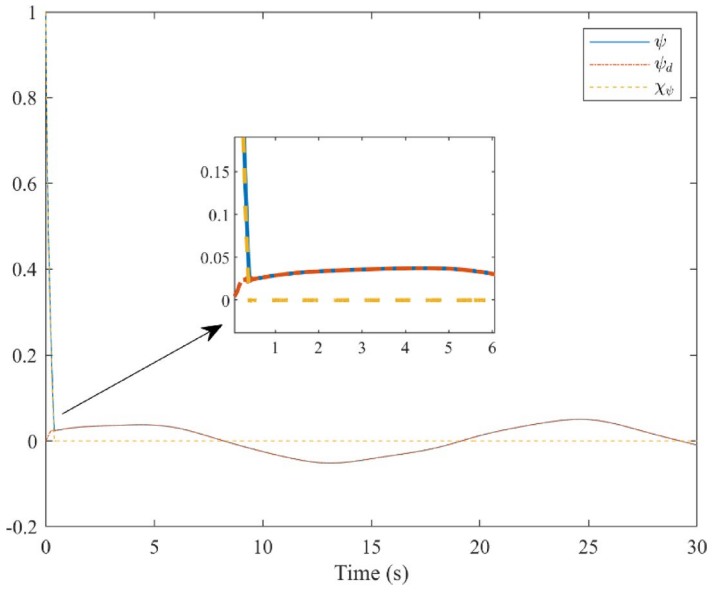

Fig. 11. The trajectory of ψ, ψ_d, χ_ψ.

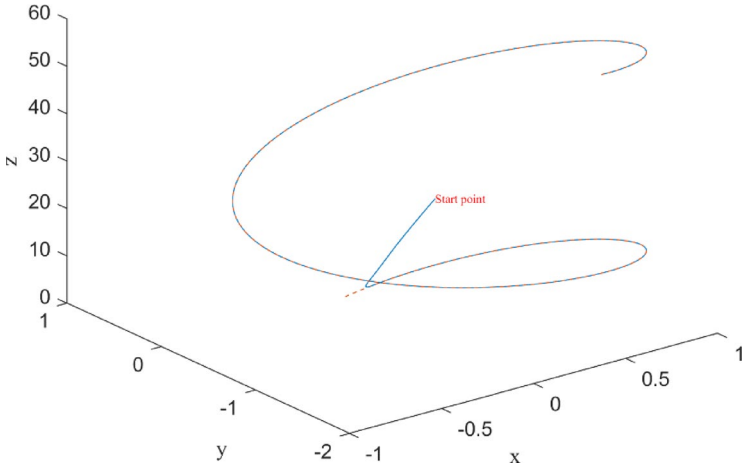

Fig. 12. 3-D spiral trajectory.

Performance index	MFC[32]	FTC[33]	Proposed PT-SMC
ITAE (position)	12.45	8.91	5.23
ISE (attitude)	6.78	4.12	2.05
Total control effort	1550	1850	2100
Max settling time (s)	∞	6.5 (theory)	4.38 (predefined)

Table 2. Performance comparison of different controllers.

Data availability

All data generated or analysed during this study are included in this published article.

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Author contributions

L.H. and J.Z. wrote the main manuscript text, Z.Y., X.W. and C.S.C. prepared simulation. All authors reviewed the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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